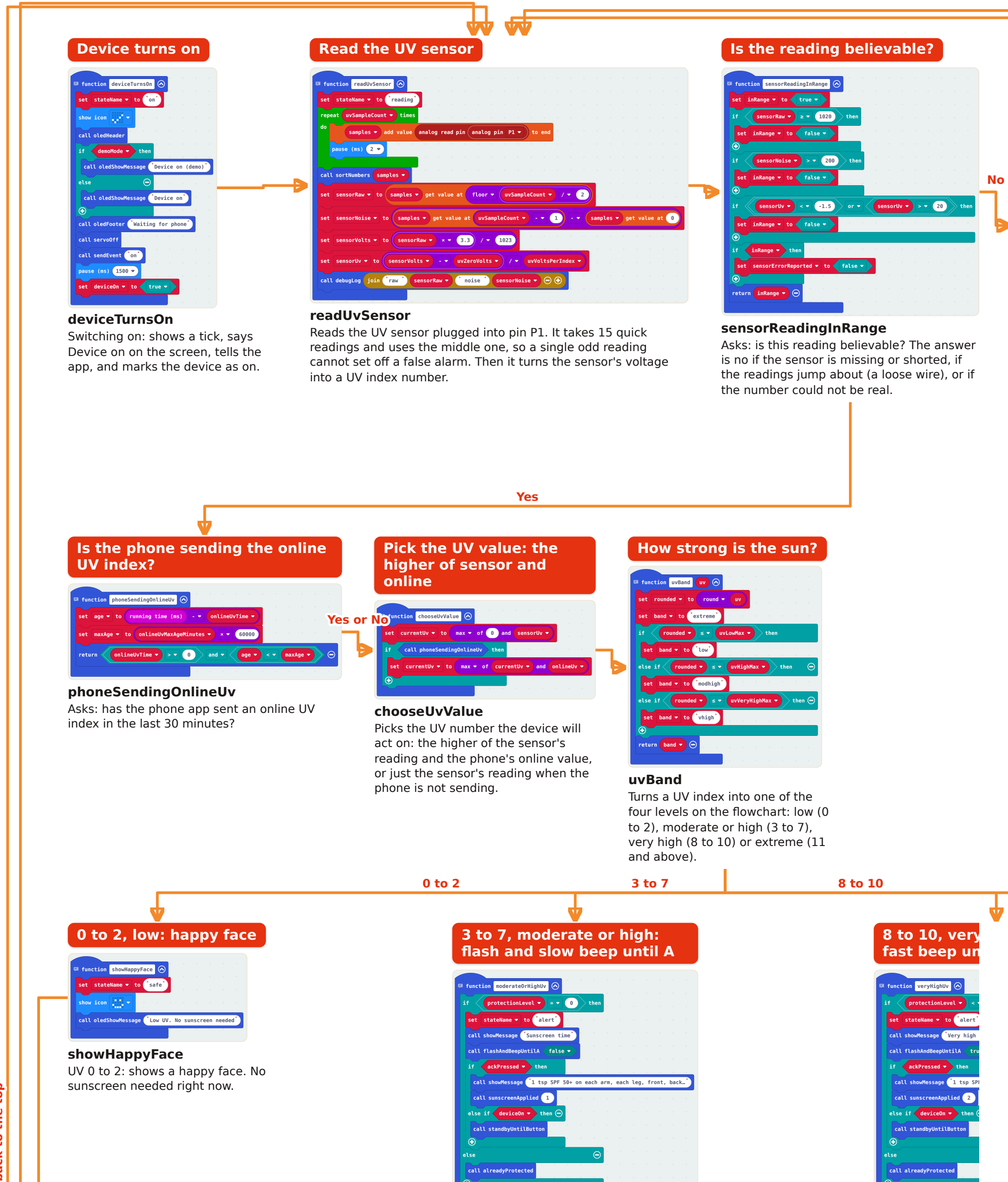


Sunburn device: the program or

The Sunburn Flowchart, built from the real blocks of the program, with what each one does

The Sunburn device is a wearable built on a BBC micro:bit, a small computer with 25 lights and two buttons. It measures the sun's UV where you are through a phone, and then flashes, beeps or taps your arm until you put sunscreen on. Two hours later it reminds you to built from 76 coloured blocks. Every one of them is on these sheets, with a note saying what it does.

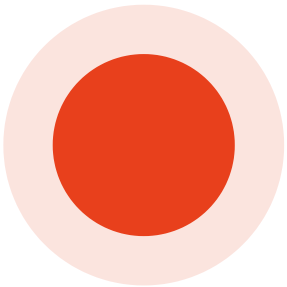
Follow the arrows. Each red label is a box of the flowchart; the coloured picture under it is the actual block of code that does that job, exactly as it looks in the Ma forever block runs this plan round and round, all day.



on the wrist

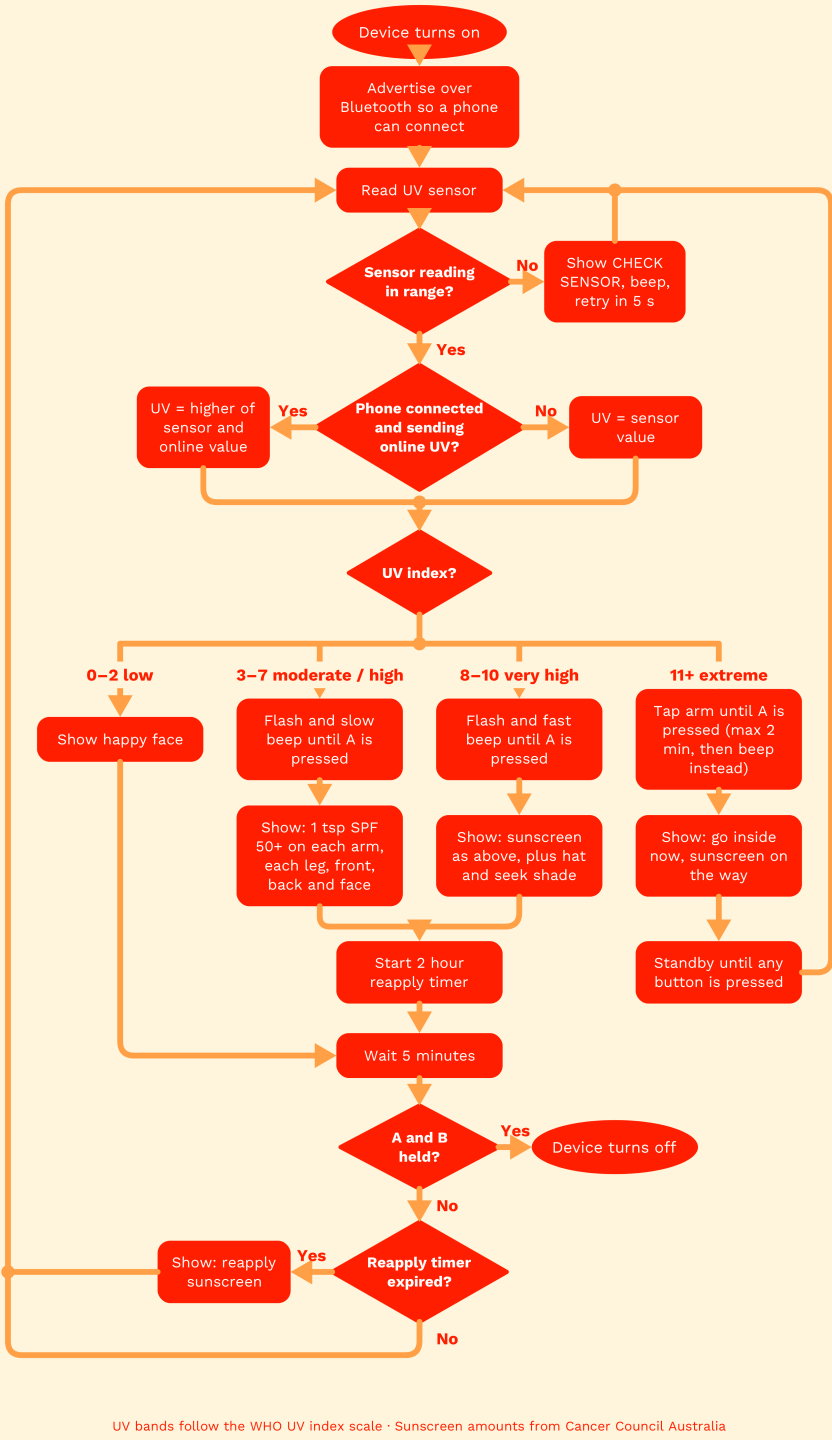
with a sensor, checks the online UV index for
o reapply. The program that does all this is

akeCode editor, and the note says what it does. The



The Sunburn Flowchart

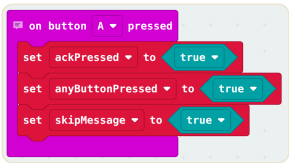
Sunburn Flowchart



The plan the device follows. Every box on it is a block on the left, with the same name.

The two buttons

A: done

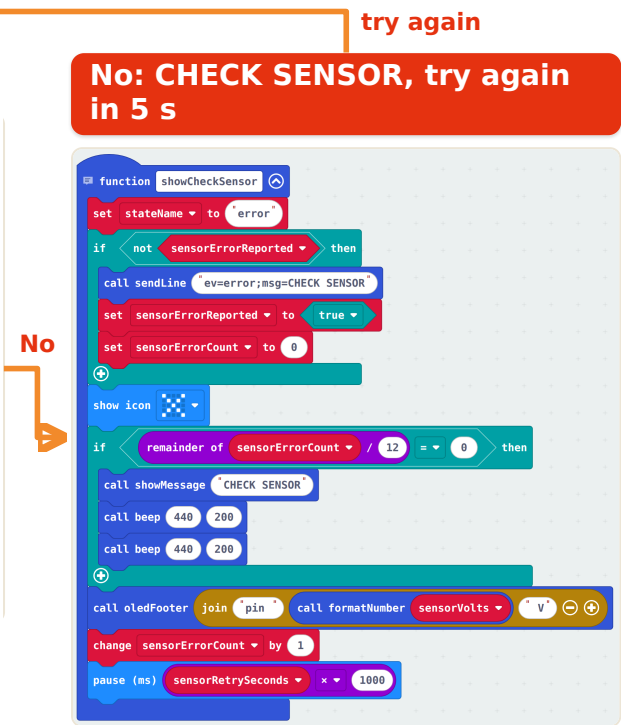


on button A pressed

Pressing A means done: sunscreen is on, or I am going inside. It also wakes the device, skips a long message, and during a wait shows the UV number and the countdown.

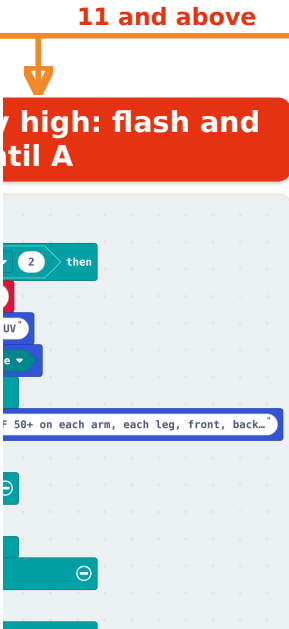
Pressing A and B together turns the device off, or on again (see the flowchart).

The whole program

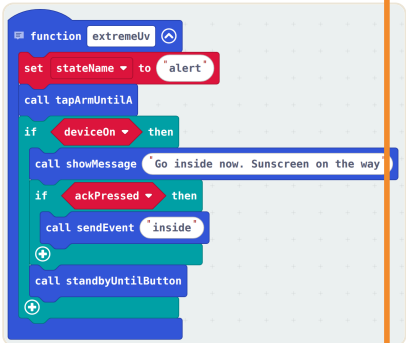


showCheckSensor

When the reading is not believable: shows a cross, and the first time scrolls CHECK SENSOR and beeps twice, then again about once a minute while the fault lasts. It tells the app once, and tries again every 5 seconds. A faulty sensor is never treated as safe.

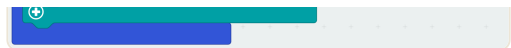


11 and above, extreme: tap the arm until A



extremeUV

UV 11 or more: taps the arm until A is pressed (after 2 minutes it beeps)



moderateOrHighUv

UV 3 to 7: flashes and beeps slowly until A is pressed, then shows how much sunscreen to put on and starts the 2 hour timer. If sunscreen is already on, it just shows a tick and the time left. If nobody presses A within 10 minutes, the device assumes it is not being worn and goes to sleep.



veryHighUv

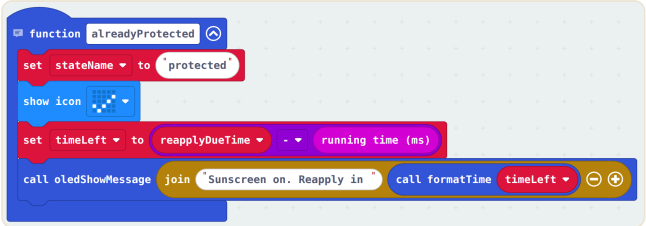
UV 8 to 10: flashes and beeps until A is pressed, then shows sunscreen advice and starts the 2 hour timer. If nobody presses A within 10 minutes, the device assumes it is not being worn and goes to sleep.

sunscreen already on

A pressed

A pressed

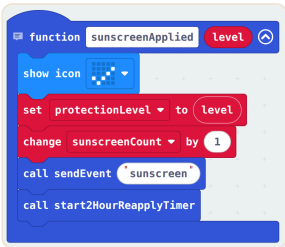
Sunscreen already on: just a tick



alreadyProtected

Sunscreen is already on, so there is no alert: just a tick and the time until it needs reapplying.

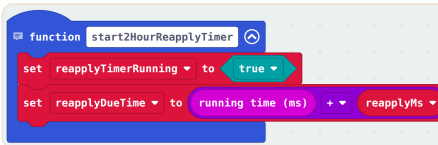
A pressed: sunscreen is on



sunscreenApplied

A was pressed after a sunscreen alert: remembers which advice was given, counts one more application, tells the app, and starts the timer.

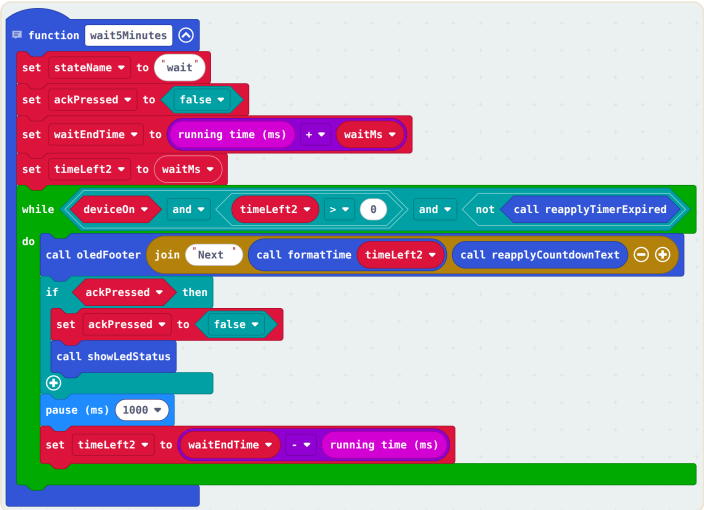
Start the 2 hour reapply timer



start2HourReapplyTimer

Starts the 2 hour countdown to the next sunscreen.

Wait 5 minutes



wait5Minutes

Waits 5 minutes before the next check. The wait ends early if the device is turned off or the sunscreen timer runs out. Pressing A during the wait shows the UV number and the countdown.

A and B held?

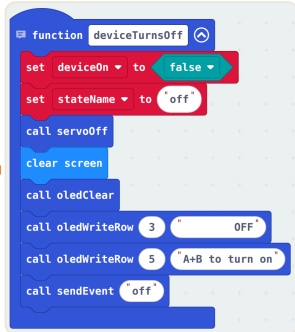


on button A+B pressed

Pressing A and B together turns the device off, or on again.

Yes

Yes: device turns off



deviceTurnsOff

Switching off: everything stops and the screen shows OFF. Bluetooth stays on so the app can switch it back on.

No

Is it time for more sunscreen?

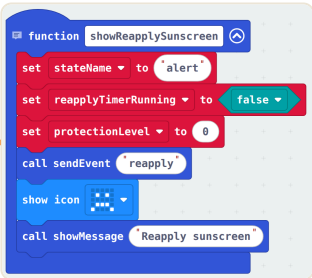


reapplyTimerExpired

Asks: have the 2 hours since the last sunscreen passed?

Yes

Yes: show Reapply sunscreen

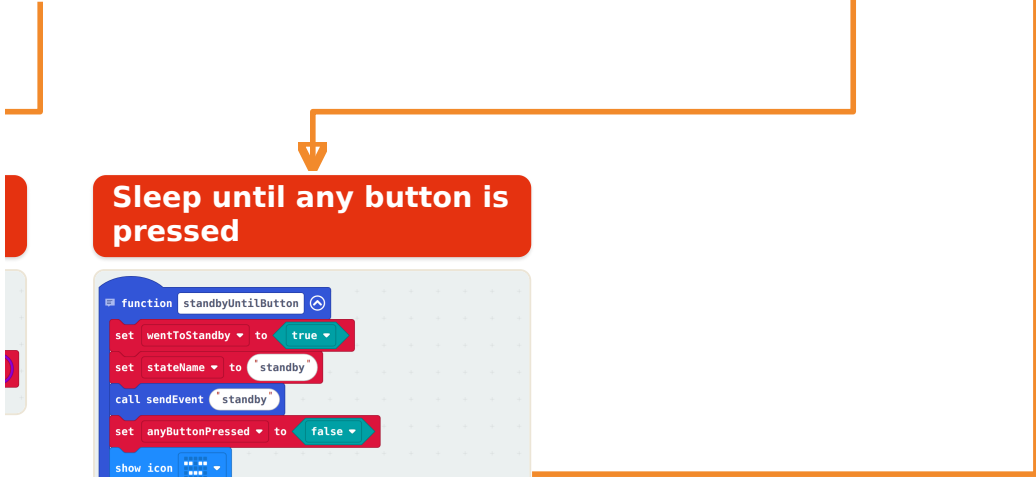


showReapplySunscreen

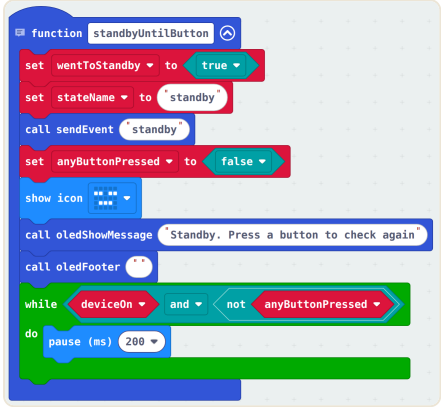
Shows Reapply sunscreen and clears the timer, so the next trip round the flowchart alerts again if the sun is still strong.



s and beeps quickly then shows the plus hat and shade, and imer. It alerts again if only had the moderate



Sleep until any button is pressed



standbyUntilButton
Sleep: waits for any button, or done from the app. Pressing A and B together still turns the device off. Afterwards it goes straight back to reading the sensor.

is pressed (after 2 minutes it beeps instead), shows Go inside now, then sleeps until a button is pressed.



All 76 blocks in the editor, in nine sections. The flowchart on this sheet is sections 1 to 4; sections 5 to 9 are the helpers: screen, sound, Bluetooth, maths and the optional screen.

What the block colours mean

- | | |
|-------------------------------|-----------------------------|
| Variables: remember a value | Text |
| Functions: a step with a name | Arrays: lists |
| Basic: the lights, pauses | Music: beeps |
| Input: buttons and time | Pins: sensor, motor, screen |
| Loops: repeat | Bluetooth: the phone |
| Logic: if, compare | Serial: the USB cable |
| Maths | Control: timing |

Words used here

- UV index** How strong the sun's burning rays are, from 0 (none) to 11 and above (extreme). Protection is needed from 3 upwards, even on cool or cloudy days.
- micro:bit** The small computer the program runs on. It has a 5 by 5 grid of lights, buttons A and B, and sockets called pins (P0, P1, P2) that other parts plug into.
- Block** One instruction of the program. Blocks snap together and run in order from top to bottom. A block with a name at the top (function) is a step that other blocks can call on.
- Sensor** The part that measures UV. It gives a voltage, which the program turns into a UV index number.
- Servo** A small motor with an arm. Here it taps the wearer's wrist when the sun is extreme.
- Bluetooth and USB** The two ways the wearable talks to a phone or a laptop. The phone app sends the online UV index; the wearable sends back what it is doing.
- OLED** An optional small screen. Without one, the grid of lights does the job.
- ms and Hz** ms are milliseconds, thousandths of a second. Hz sets how high a beep sounds.
- Demo timings** Much shorter waits (seconds instead of minutes and hours) so the whole day can be shown in a few minutes at a demo.
- Names in the blocks** Words like uvSampleCount are the program's own names for its settings and memory. Read them as English: uv sample count is how many readings to take.

Every one of the 76 blocks, with its note, is in the booklet on the table.